

Hanwen Zhang 张瀚文

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Seeking long-term remote research internships. Planning to apply for Fall 2028 PhD admission.

Education

Sichuan University

Sep 2024 – Present

B.Eng. in Computer Science and Technology
Top-notch Student Training Program 2.0 (Honors Class)

GPA: 3.96/4.00 • Rank: 1st in honors class

Weighted average: 93.76/100. Grades as of September 23, 2026.

Coursework and additional grade details

Compulsory-course GPA: 3.95/4.00; compulsory-course weighted average: 93.61/100.

Selected coursework: Operating System (99), Theory of Computation (98), Theory of Optimization (98), Probability Statistics (97), Numerical Computing Method (97), Introduction to Artificial Intelligence (96), Discrete Mathematics (94), Introduction to Multimodal Learning (90).

Research Interests

Visual generation and 3D reconstruction, with a longer-term interest in generative world models for spatial reasoning and action.

Research Experience

Bone-to-Shape: Single-Image Animal Reconstruction

2026

Project Lead • Sichuan University

Provincial Undergraduate Innovation Training Program

- Lead a project that generates orbit-view video from a single animal image and reconstructs a 3D Gaussian Splatting (3DGS) representation.
- Explore anatomical and body-shape priors to constrain reconstructed geometry.

Project overview →

Project title: Bone-to-Shape: Anatomy-Constrained 3D Reconstruction for Veterinary Applications (骨影生形).

Passed the college midterm review. Related patent application details are listed under Intellectual Property.

City University of Hong Kong

Jul 15 – Sep 1, 2026

Research Intern • Department of Data Science

Supervisor: Xinke Li

- Investigated generative visual priors for 3D reconstruction using TRELIS.2.
- Contributed to ground-truth calibration and geometric registration for FARO laser scanning in complex physical environments.

Deep-Hole Guardian (深孔卫士)

2026

Team Member

University-Level Undergraduate Innovation Training Program

- Participated in a project on embodied perception for deep blind-hole defect identification and control with a SCARA robot.
 - Passed the college midterm review.
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Sparse-View Generation and Reconstruction

Research Project

- Study sparse-view 3D generation and reconstruction using TRELIS and GenRecon.
 - Investigate how generative priors complete unseen structure while preserving observed evidence.
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Kernel-Based Outlier Detection

Published 2026

Research Collaboration • Sichuan University

- Contributed to method design and formal analysis for kernel self-representation learning based fuzzy neighborhood outlier detection.
 - Third author of the resulting article in International Journal of Approximate Reasoning.
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Publications

IJAR 2026

Kernel self-representation learning based fuzzy neighborhood outlier detection

Yijia Chen, Ze Zhang, Hanwen Zhang, Jiaqi Yang, Bingye Zhou, Zhong Yuan

International Journal of Approximate Reasoning, vol. 198, article 109769, 2026. Third author; contributed to method design and formal analysis.

Paper

Patents & Software Copyrights

Animal 3D Reconstruction Using Body-Shape Priors

基于体型先验的动物三维重建方法

First-listed inventor • Applicant: Sichuan University

Chinese invention patent application 202611403381.7

Filed September 10, 2026

Passed preliminary examination on September 22, 2026. Pending publication; not yet granted.

Granular Network Threat Detection System Based on Subspace Learning

基于子空间学习的粒网络威胁检测系统 V1.0

Joint copyright holder

Chinese software copyright 2025SR1158898

Registered July 3, 2025

Honors

- **First-Class Single-Category Scholarship (单项一等奖学金)**, Sichuan University, 2025–2026 academic year.
- **Third-Class Comprehensive Scholarship (综合三等奖学金)**, Sichuan University, 2024–2025 academic year.
- **Third Prize, Sichuan Division**, 17th Chinese Mathematics Competition for College Students, Non-Mathematics Category A, 2025.
- **Outstanding Undergraduate Student Cadre**, Sichuan University, 2024–2025 academic year.
- **Provincial Third Prize, Ascend AI Track**, Huawei ICT Competition, 2024.

COMPETITIONS & ASSESSMENTS

- **2026 MCM**: Successful Participant.
- **CCF CSP**: highest score 250/500, March 30, 2025 (top 20.6% in that sitting); top 11.23% in the May 31, 2026 sitting.

Software & Engineering Projects

Paper Refiner

Multi-agent LaTeX paper revision system

Designed and implemented the workflow and orchestrator, including structured patch-based editing and reproducible runs with configurations and logs.

Python • LaTeX • JSON patches

[GitHub](#) [Details](#)

Smart POS System

Checkout application

Worked on application refactoring, asynchronous database access, UI components, and error handling.

C++ • Qt

[GitHub](#) [Details](#)

Bouncing Birds

Physics game with save/load and replay

Worked on gameplay, collision handling, saving and loading game state, and replay functionality.

C++ • SFML

[GitHub](#) [Details](#)

Granular Network Threat Detection System

2025

Network threat detection using subspace learning and granular computing

Co-developed the system; joint holder of its registered software copyright.

Subspace learning • Granular computing

Details

Activities & Service

ACADEMIC EXCHANGE

- **UNSW Business School**, “Discovery: Foundations and Applications,” July 20–31, 2026. Completed a 30-hour in-person program; led development of a Hackathon prototype combining financial candlestick analysis and multi-agent collaboration.
- **CNCC 2025**, Harbin, October 22–25, 2025. Conference participant.

LEADERSHIP & SERVICE

- **President**, Sichuan University AI Club; previously Vice President.
 - **League Branch Secretary**, 2024 Honors Class, College of Computer Science; helped organize the class’ s successful “Jiang Jie Class” application.
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Skills

- **Programming:** C/C++, Python.
- **ML / Data:** PyTorch, scikit-learn.
- **Tools:** Linux, Git, Docker, LaTeX.
- **Languages:** Chinese (native); English (CET-6: 538).